Double-click (or enter) to edit

What is Matplotlib?

Matplotlib is a popular Python library used for data visualization. It helps us create different types of graphs and charts to represent data in a simple and attractive way.

Developed by: John D. Hunter (2003)

Main Module: matplotlib.pyplot

Features of Matplotlib Easy to create graphs and charts Supports 2D plotting Can create simple 3D plots Highly customizable Works well with NumPy and Pandas Can save graphs in different image formats (PNG, JPG, PDF, SVG) Used in data science, machine learning, business analytics, and scientific research

Advantages of Matplotlib

Free and open source Easy to learn High-quality visualizations Many chart types available Integrates with NumPy and Pandas Supports customization of colors, labels, titles, and styles

Disadvantages of Matplotlib

More code is needed compared to some modern libraries Complex graphs can take longer to create Not ideal for highly interactive dashboards

Installation

```
pip install matplotlib
```

```
Requirement already satisfied: matplotlib in /usr/local/lib/python3.12/dist-pack
Requirement already satisfied: contourpy>=1.0.1 in /usr/local/lib/python3.12/dis
Requirement already satisfied: cycler>=0.10 in /usr/local/lib/python3.12/dist-pa
Requirement already satisfied: fonttools>=4.22.0 in /usr/local/lib/python3.12/di
Requirement already satisfied: kiwisolver>=1.3.1 in /usr/local/lib/python3.12/di
Requirement already satisfied: numpy>=1.23 in /usr/local/lib/python3.12/dist-pac
Requirement already satisfied: packaging>=20.0 in /usr/local/lib/python3.12/dist
Requirement already satisfied: pillow>=8 in /usr/local/lib/python3.12/dist-packa
Requirement already satisfied: pyparsing>=2.3.1 in /usr/local/lib/python3.12/dis
Requirement already satisfied: python-dateutil>=2.7 in /usr/local/lib/python3.12
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-packag
```

Import Matplotlib

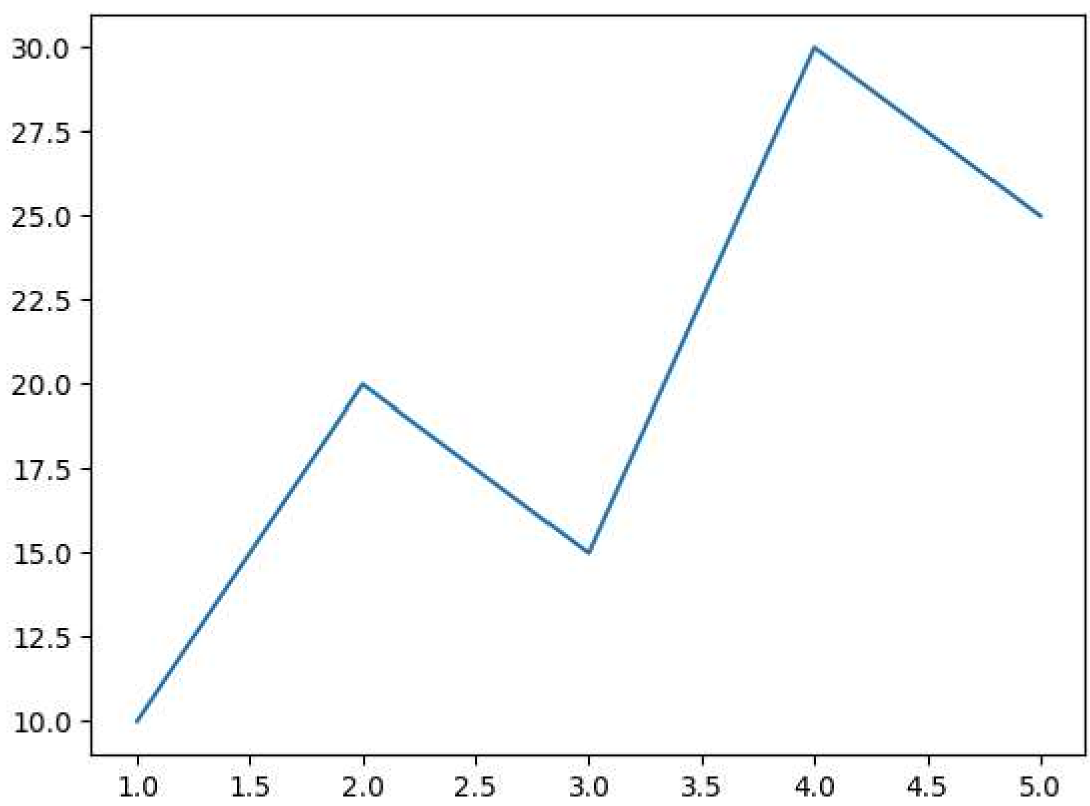
```
import matplotlib.pyplot as plt
```

Import with NumPy

```
import numpy as np
import matplotlib.pyplot as plt
```

Basic Line Plot

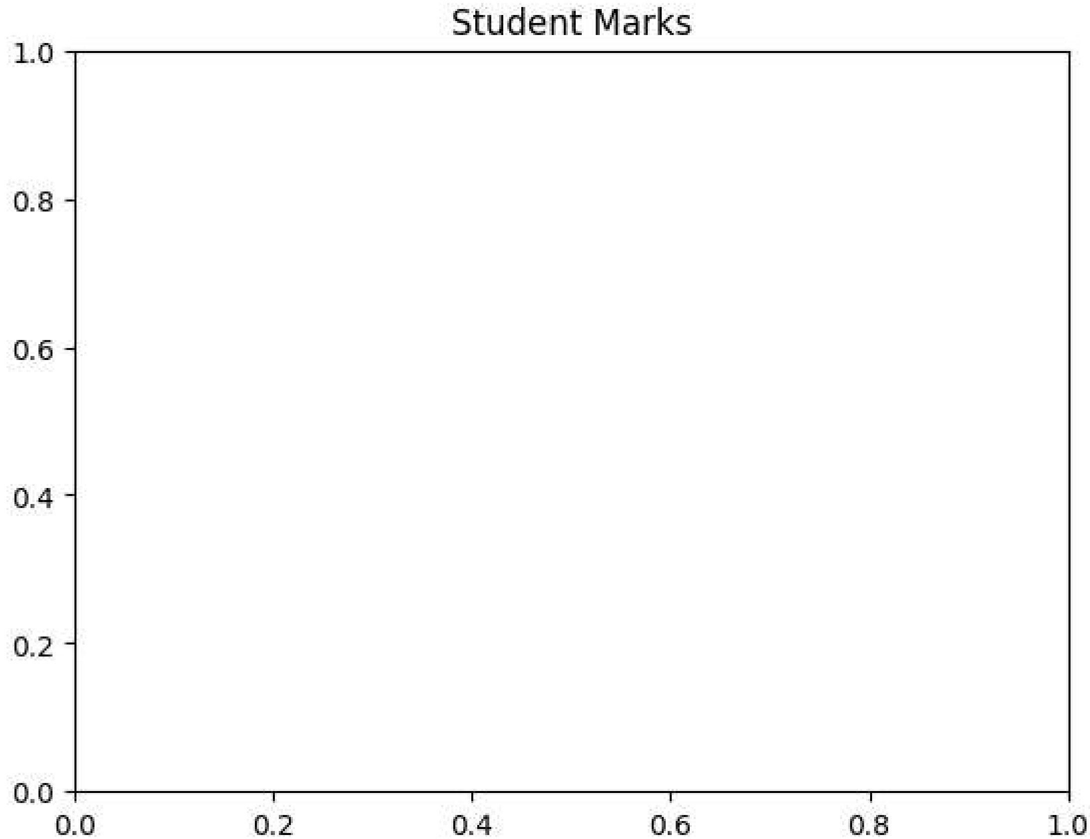
```
import matplotlib.pyplot as plt

x=[1,2,3,4,5]
y=[10,20,15,30,25]

plt.plot(x,y)
plt.show()
```



Adding Title

```
plt.title("Student Marks")
```

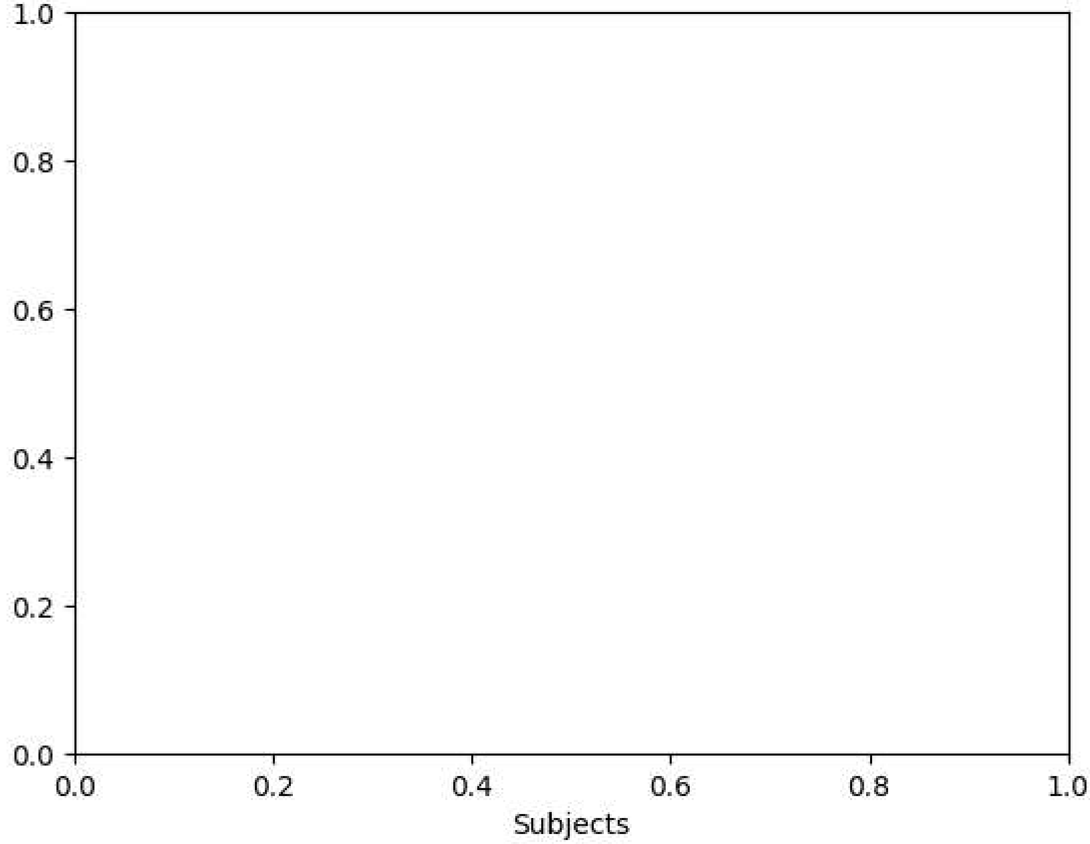
```
Text(0.5, 1.0, 'Student Marks')
```



X-axis Label

```
plt.xlabel("Subjects")
```

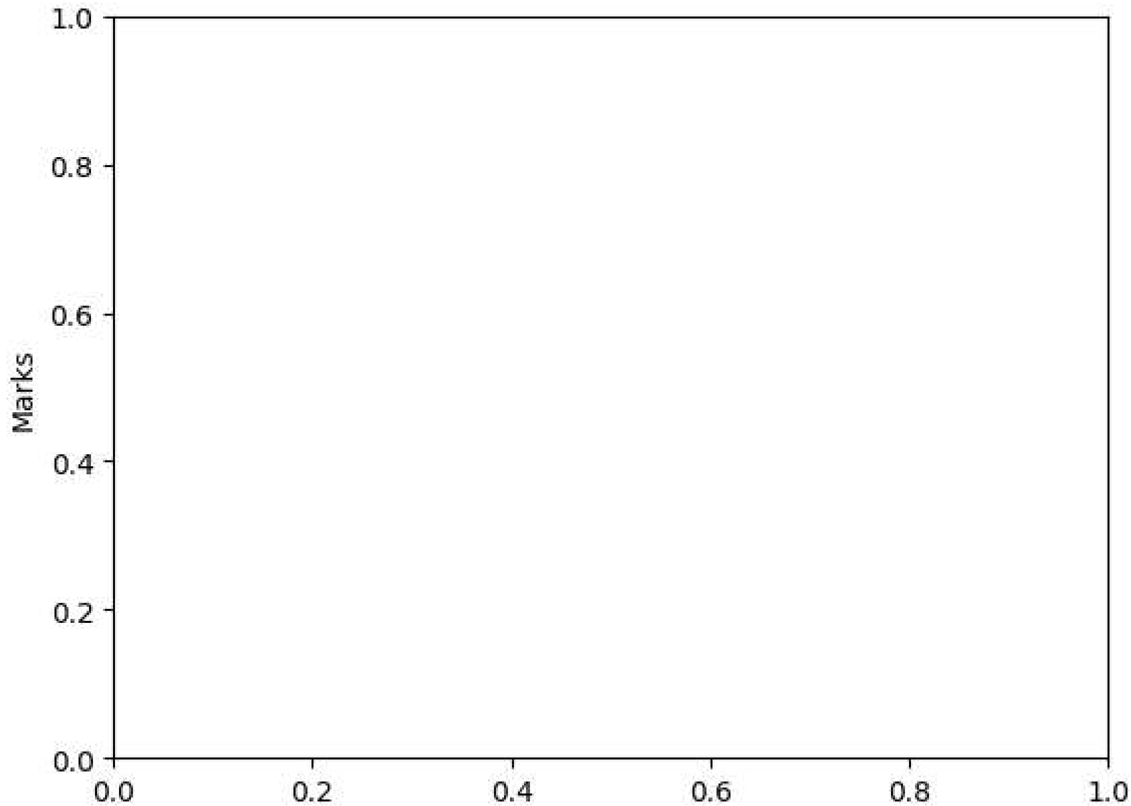
```
Text(0.5, 0, 'Subjects')
```



Y-axis Label

```
plt.ylabel("Marks")
```

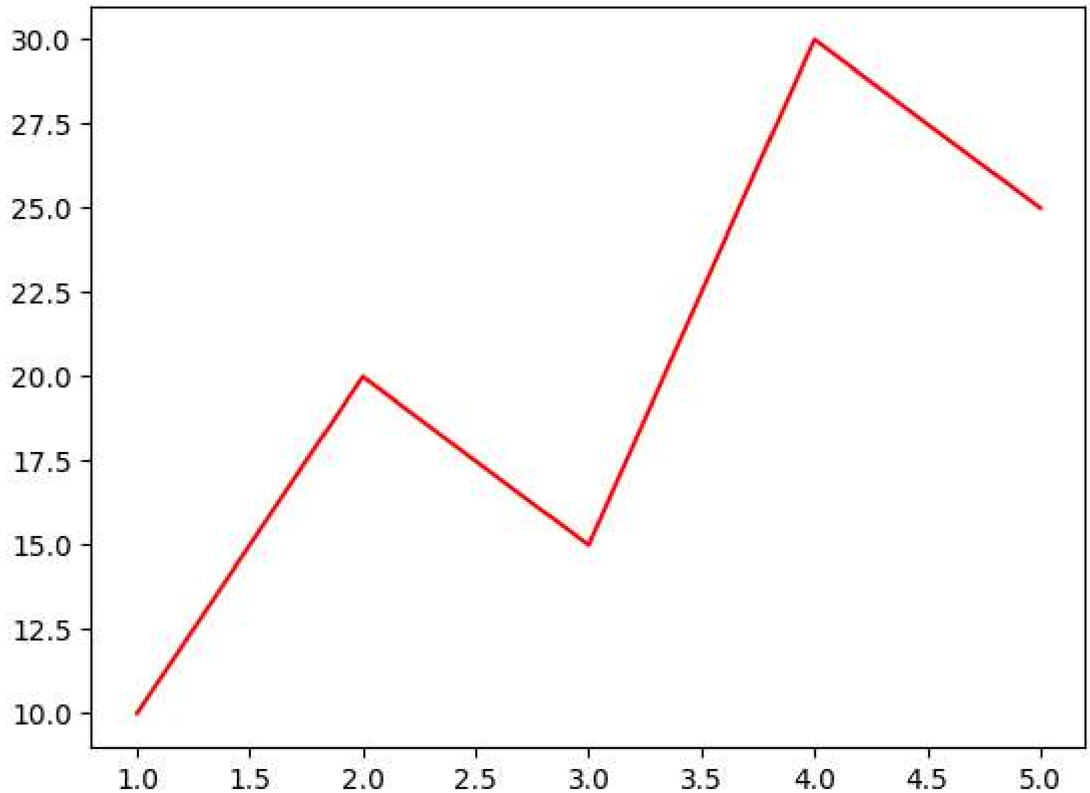
```
Text(0, 0.5, 'Marks')
```



Line Color

```
plt.plot(x,y,color="red")
```

```
[&lt;matplotlib.lines.Line2D at 0x7a93df5e6a80&gt;]
```

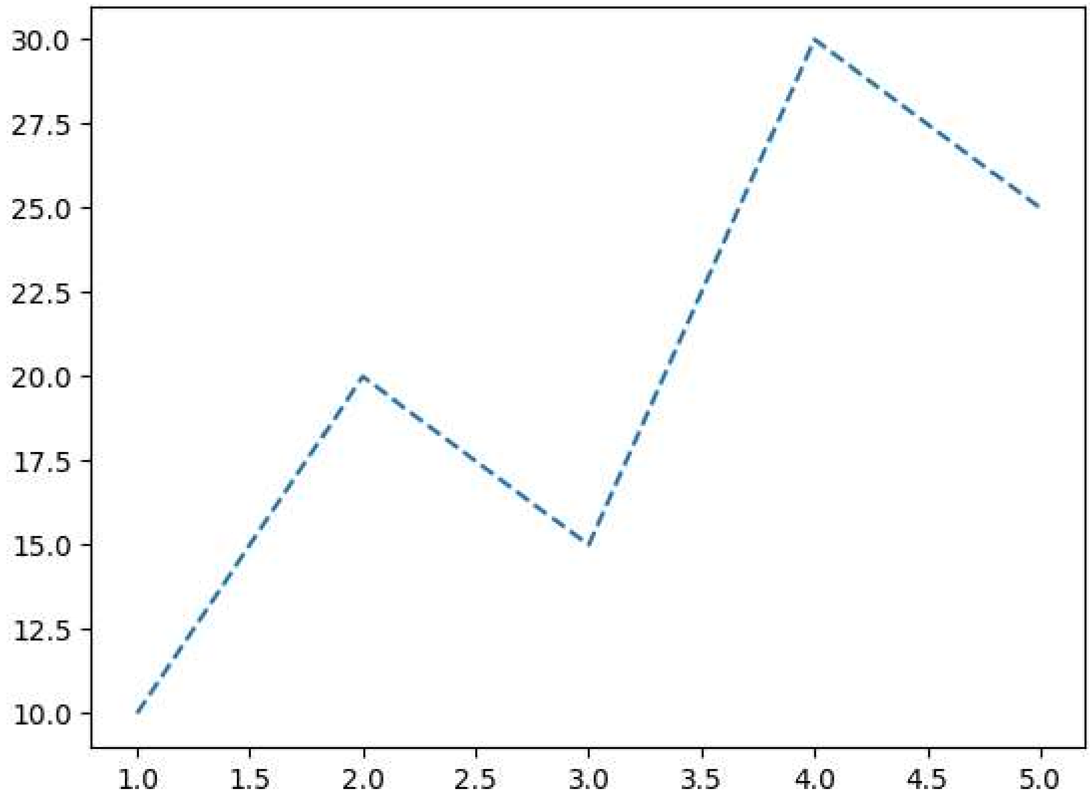


Other colors

Line Style

```
plt.plot(x,y,linestyle="--")
```

```
[<matplotlib.lines.Line2D at 0x7a93dd3350d0>]
```

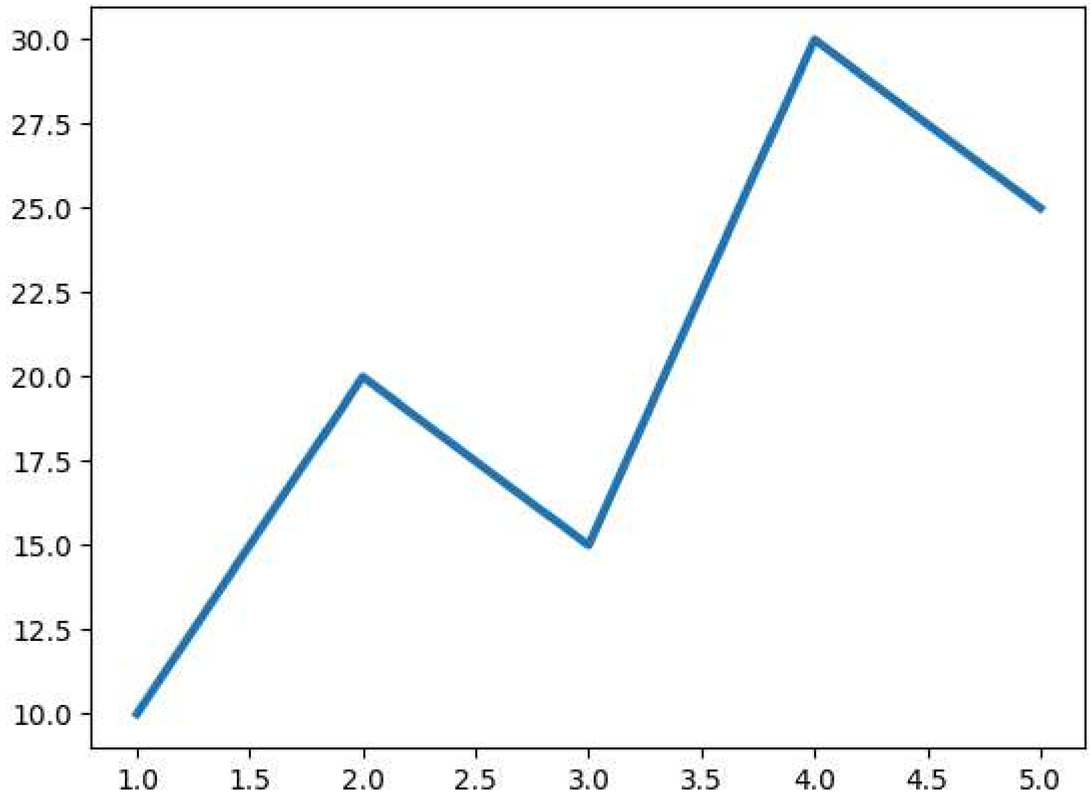


Styles

Line Width

```
plt.plot(x,y,linewidth=3)
```

```
[<matplotlib.lines.Line2D at 0x7a93dcc5fc50>]
```



Marker

plt.plot(x,y,marker="o")

Common markers

Grid

```
plt.grid(True)
```

Legend

```
plt.plot(x,y,label="Sales")
plt.legend()
```

```
<matplotlib.legend.Legend at 0x7a93dfea7fe0>
```

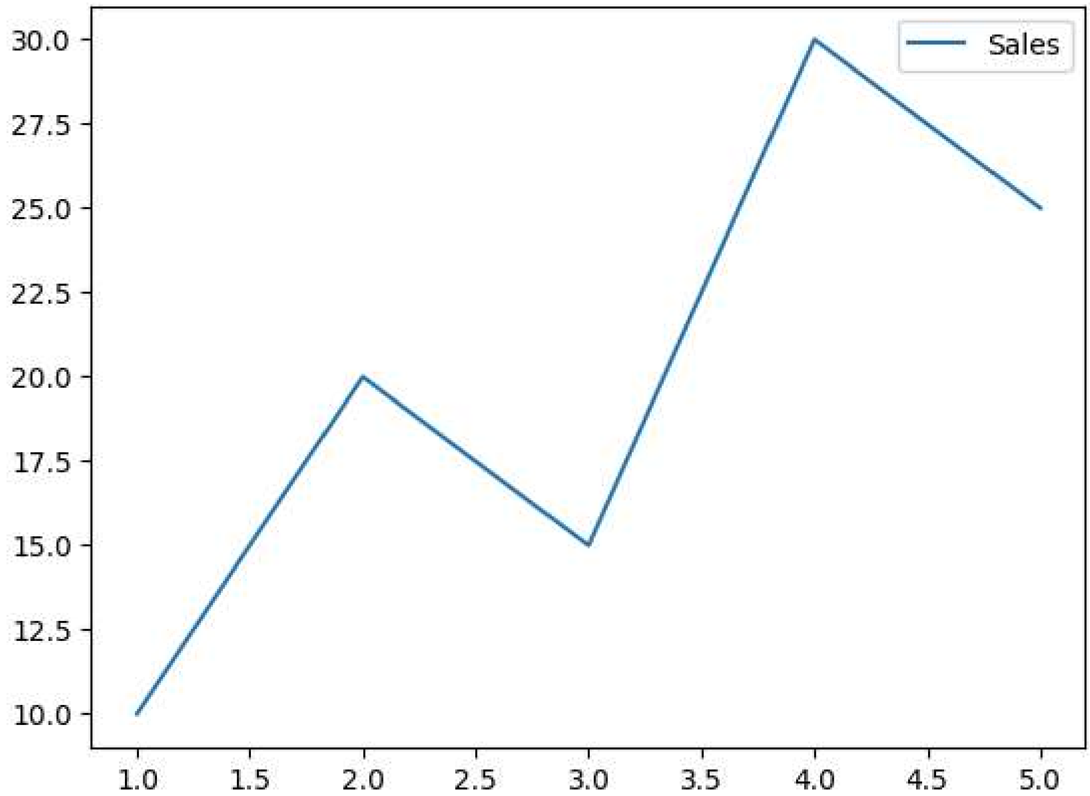


Figure Size

```
plt.figure(figsize=(8,5))
```

```
<Figure size 800x500 with 0 Axes>
<Figure size 800x500 with 0 Axes>
```

Save Figure

```
plt.savefig("graph.png")
```

```
<Figure size 640x480 with 0 Axes>
```

Show Graph

```
plt.show()
```

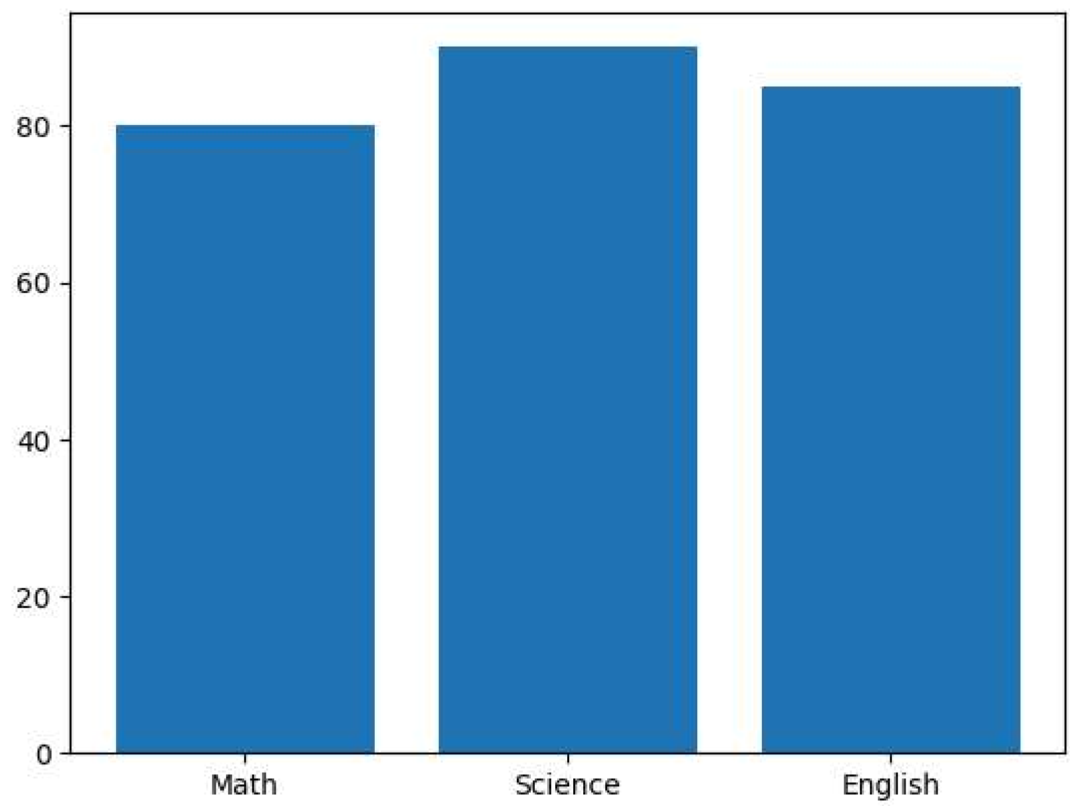
Types of Charts

1. Line Chart

Used to show trends over time.

Example

```
import matplotlib.pyplot as plt

subjects=["Math","Science","English"]
marks=[80,90,85]

plt.bar(subjects,marks)
plt.show()
```



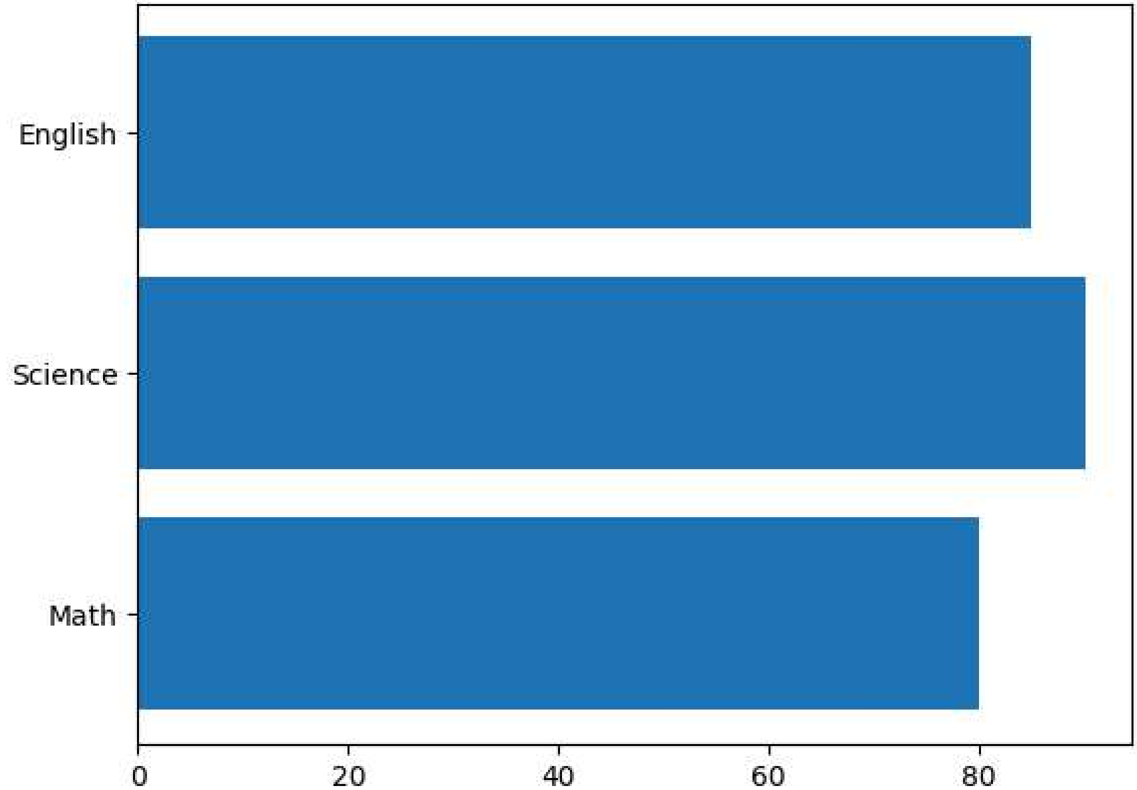
Applications

Student marks Product sales Population comparison

3. Horizontal Bar Char

```
plt.barh(subjects,marks)
```

```
<BarContainer object of 3 artists>
```

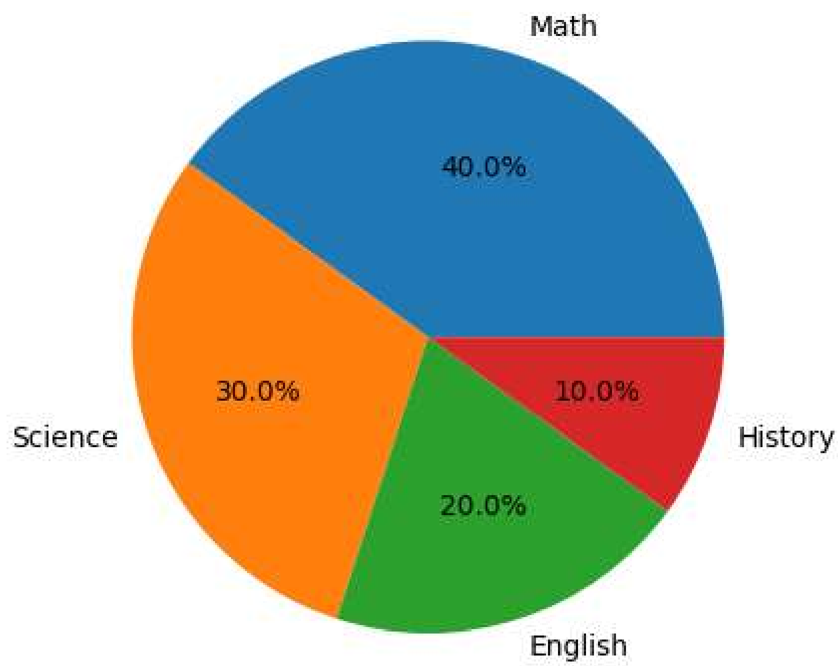


4. Pie Chart

Shows percentage distribution

Example

```
import matplotlib.pyplot as plt

marks=[40,30,20,10]
labels=["Math","Science","English","History"]

plt.pie(marks,labels=labels,autopct="%1.1f%%")
plt.show()
```



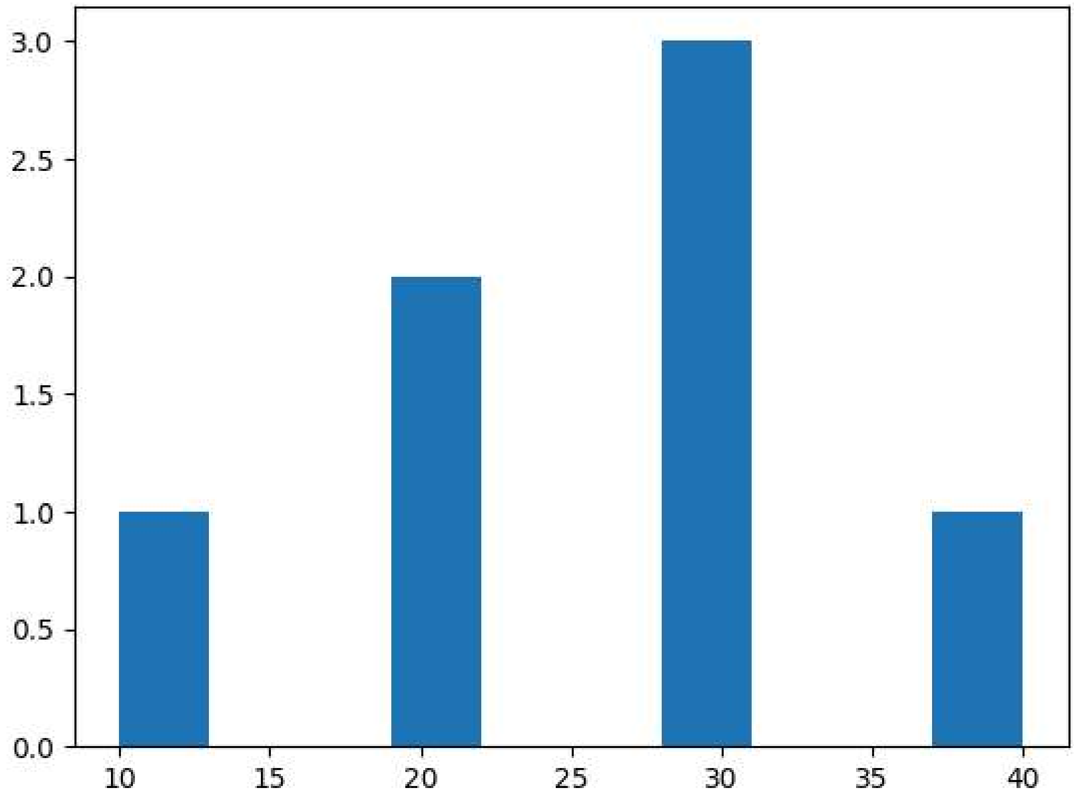
Applications

Budget distribution Market share Population percentage

5. Histogram

Example

```
import matplotlib.pyplot as plt

data=[10,20,20,30,30,30,40]

plt.hist(data)
plt.show()
```



Applications

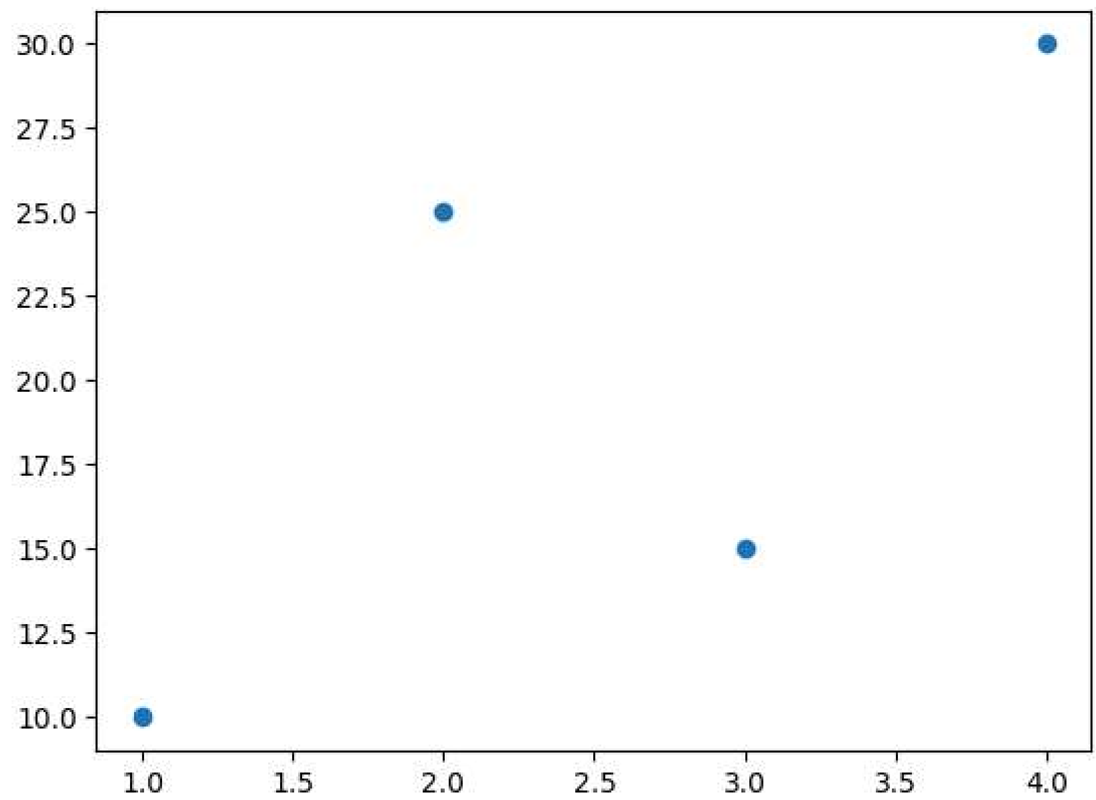
Age distribution Exam score distribution

6. Scatter Plot

Shows relationship between two variables.

Example

```
import matplotlib.pyplot as plt

x=[1,2,3,4]
y=[10,25,15,30]

plt.scatter(x,y)
plt.show()
```



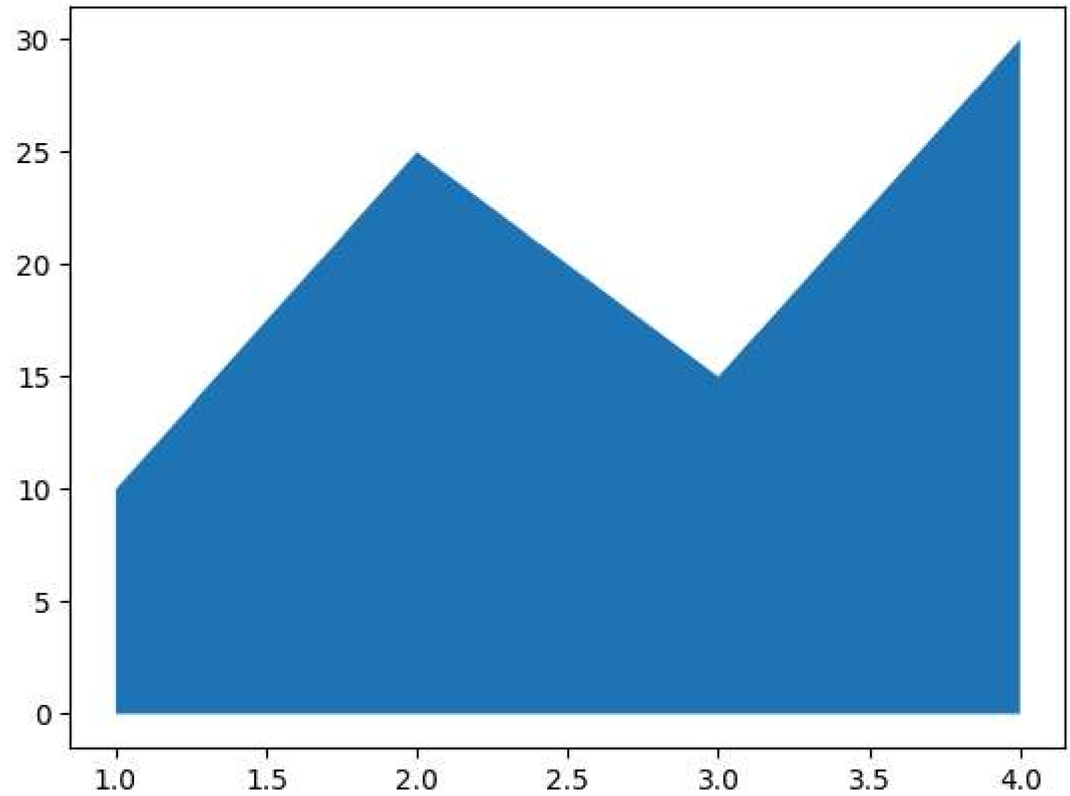
Applications

Height vs Weight Salary vs Experience

7. Area Chart

```
plt.fill_between(x,y)
```

```
<matplotlib.collections.FillBetweenPolyCollection at 0x7a93dcdb3f80>
```



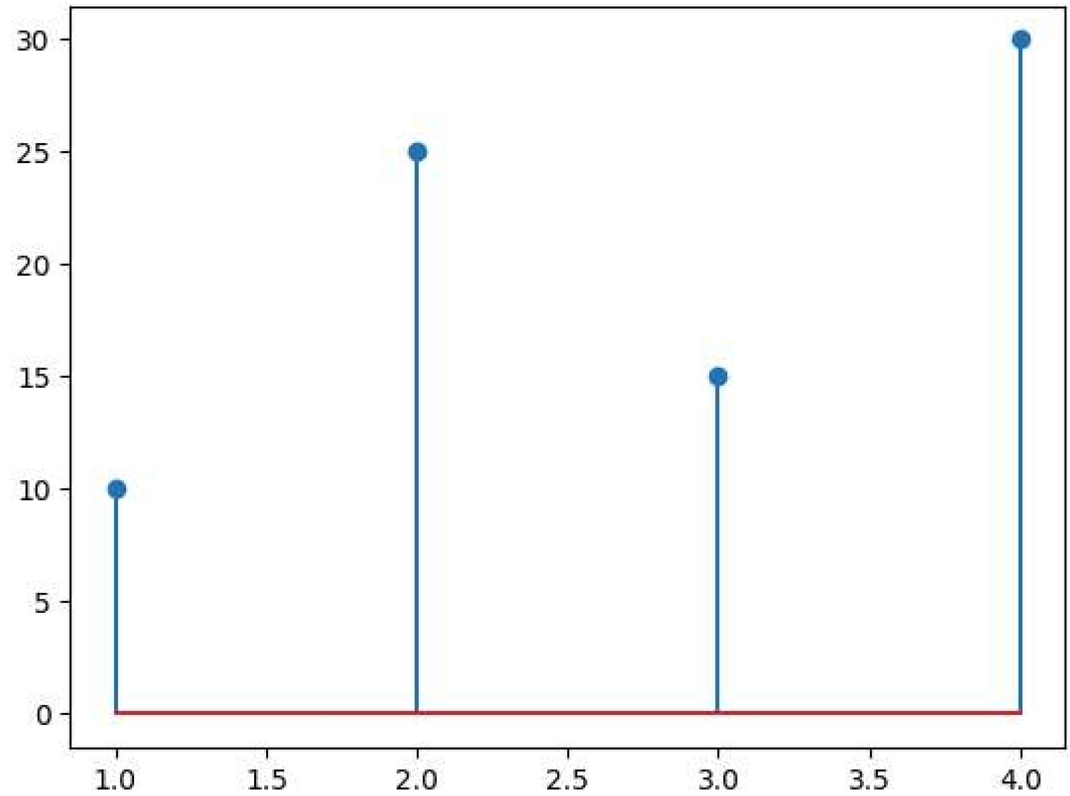
Applications

Population growth Sales growth

8. Stem Plot

```
plt.stem(x,y)
```

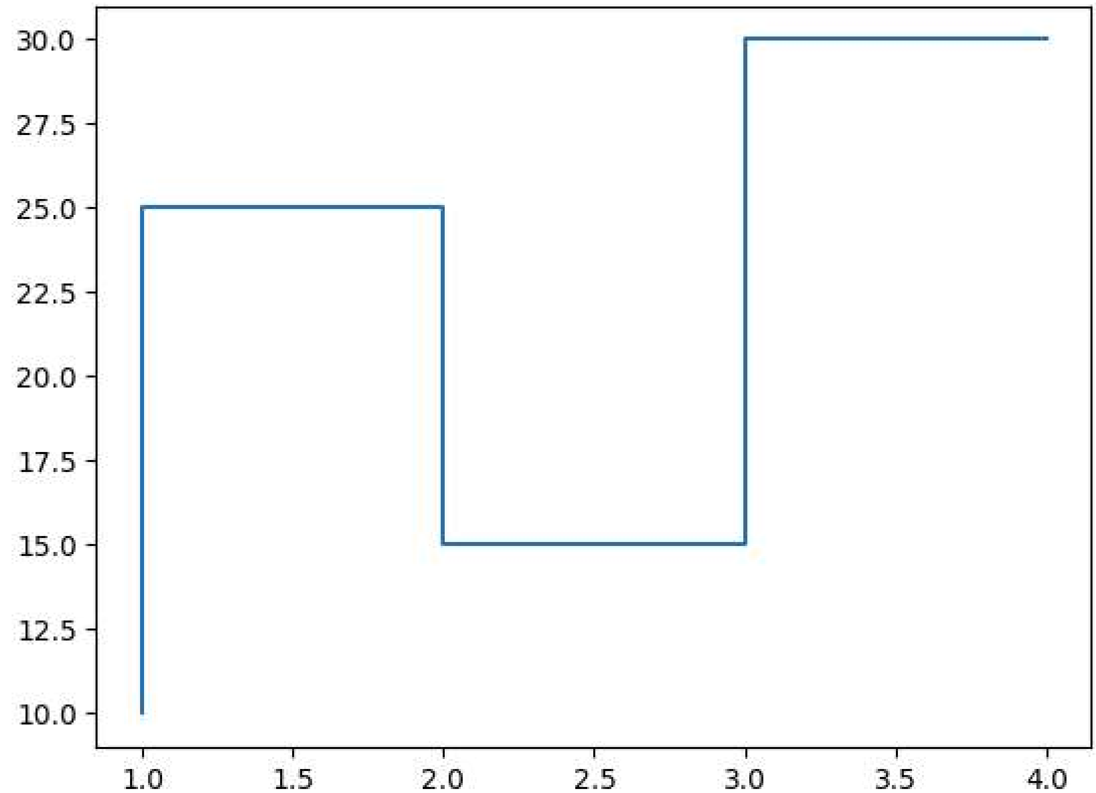
```
<StemContainer object of 3 artists>
```



9. Step Plot

```
plt.step(x,y)
```

```
[<matplotlib.lines.Line2D at 0x7a93dd20ee40>]
```

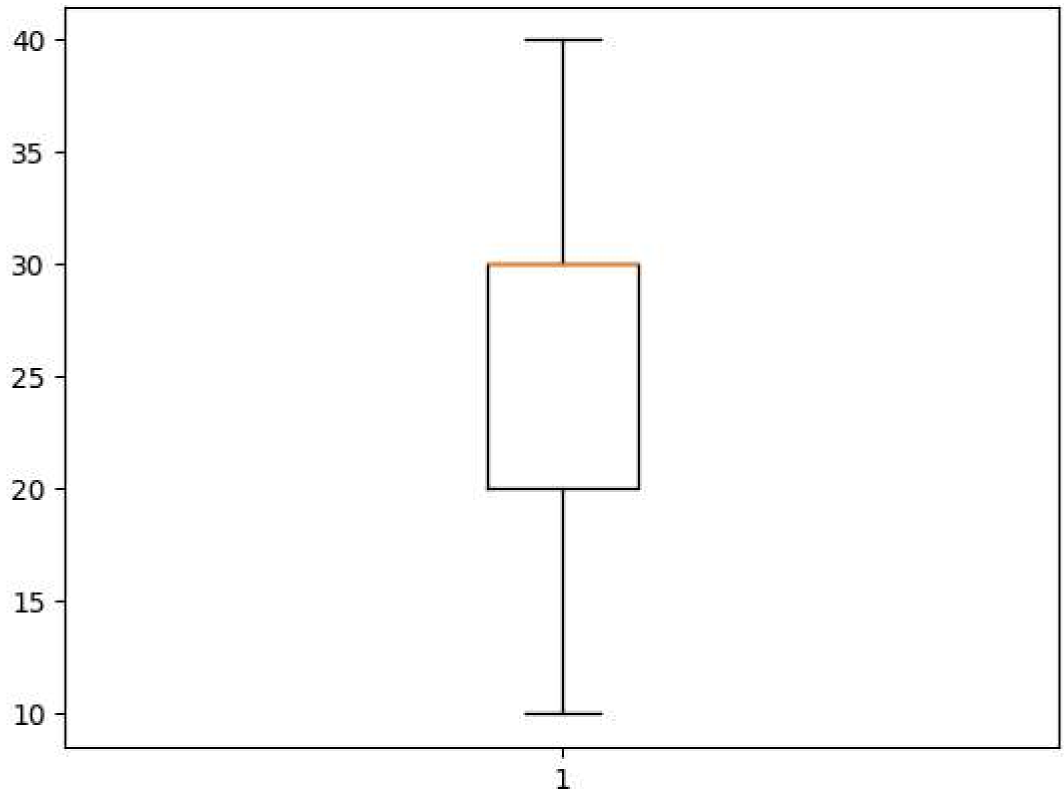


10. Box Plot

Used to identify data spread and outliers.

```
plt.boxplot(data)
```

```
{'whiskers': [<matplotlib.lines.Line2D at 0x7a93dd1dc560>,
  <matplotlib.lines.Line2D at 0x7a93dd1dc800>],
 'caps': [<matplotlib.lines.Line2D at 0x7a93dd1dcad0>,
  <matplotlib.lines.Line2D at 0x7a93dd1dce00>],
 'boxes': [<matplotlib.lines.Line2D at 0x7a93dd1dc2c0>],
 'medians': [<matplotlib.lines.Line2D at 0x7a93dd1dd0d0>],
 'fliers': [<matplotlib.lines.Line2D at 0x7a93dd1dd370>],
 'means': []}
```

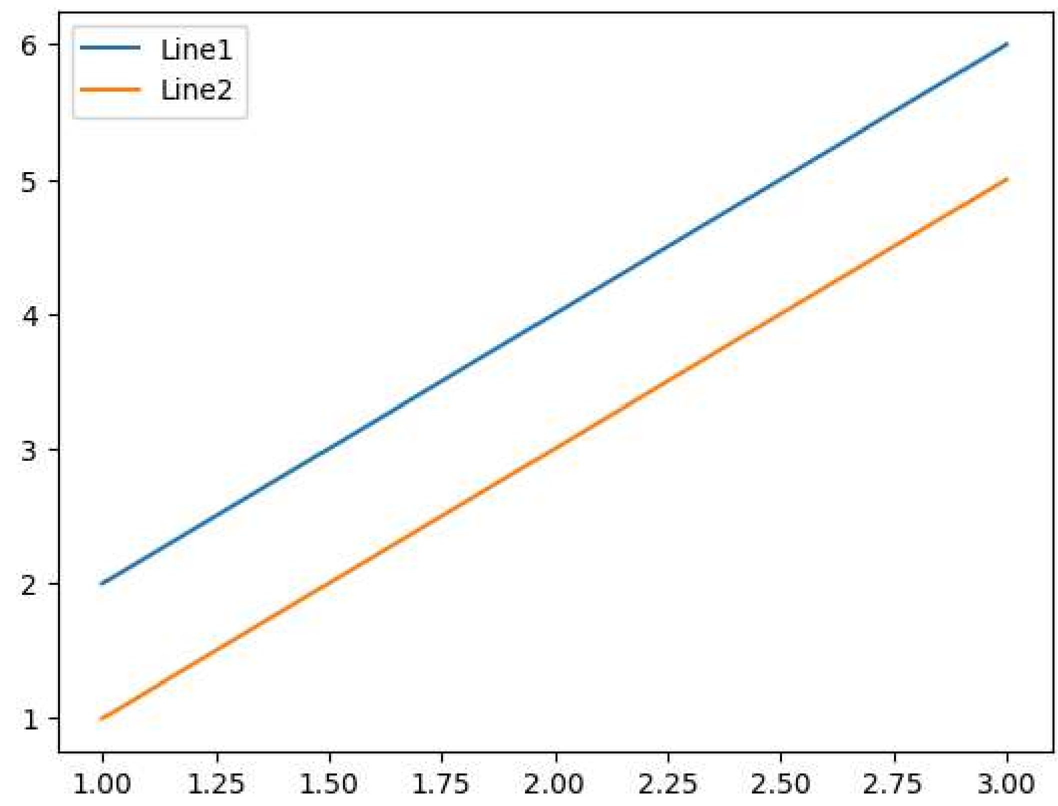
11. Error Bar Plot

```
plt.errorbar(x,y,yerr=2)
```

Multiple Lines

```
import matplotlib.pyplot as plt

x=[1,2,3]

y1=[2,4,6]
y2=[1,3,5]

plt.plot(x,y1,label="Line1")
plt.plot(x,y2,label="Line2")

plt.legend()
```

```
plt.show()
```



Axis Limits

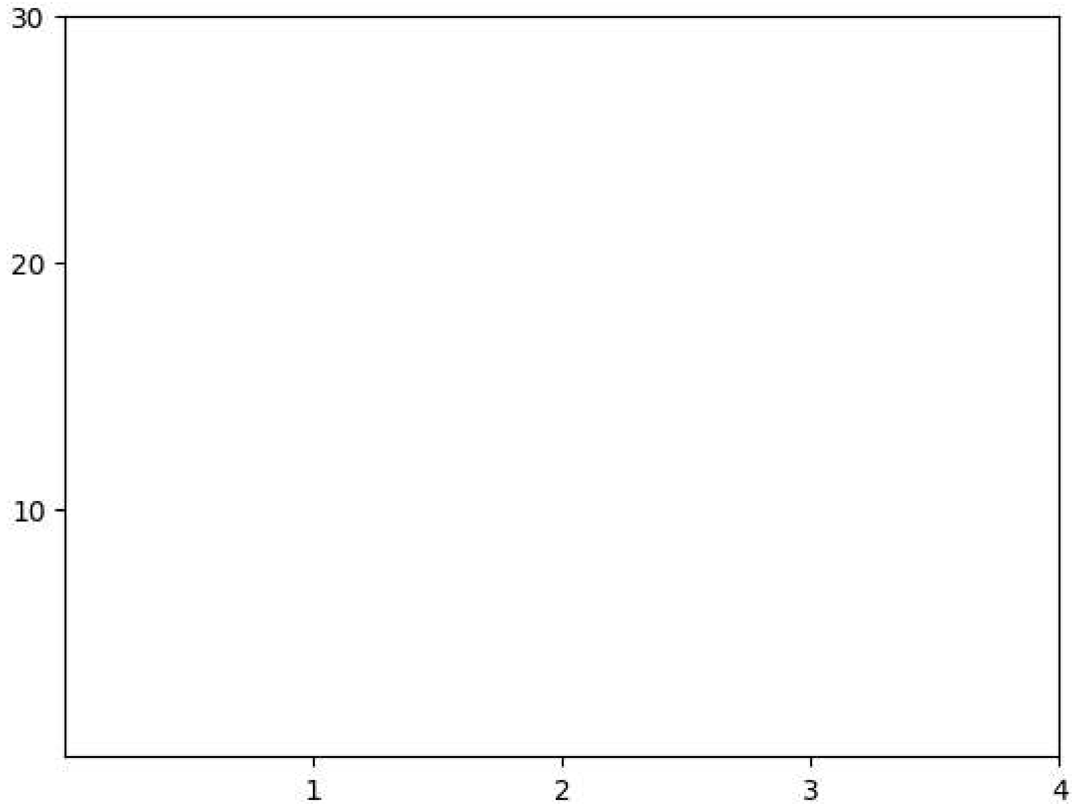
plt.xlim(0,10)

plt.ylim(0,50)

Axis Ticks

```
plt.xticks([1,2,3,4])

plt.yticks([10,20,30])
```

```
([<matplotlib.axis.YTick at 0x7a93dd273740>,
  <matplotlib.axis.YTick at 0x7a93dd23be30>,
  <matplotlib.axis.YTick at 0x7a93dcb67b30>],
 [Text(0, 10, '10'), Text(0, 20, '20'), Text(0, 30, '30')])
```



Text on Graph

```
plt.text(2,20,"Highest")
```

```
Text(2, 20, 'Highest')
```

Highest